

Tokenizer Compression Comparison

Bytes-per-token ratio: higher is better (fewer tokens per byte)

vs. GPT-2 (vocab 50 257)

		GPT-2		Ours			
Text Type	Bytes	Tokens	Ratio	Tokens	Ratio	Rel. Diff %	Winner
news	1813	398	+4.6%	592	+3.1%	-48.7%	GPT-2
korean	877	735	+1.2%	844	+1.0%	-14.8%	GPT-2
code	1259	576	+2.2%	672	+1.9%	-16.7%	GPT-2
math	1830	933	+2.0%	1107	+1.6%	-18.6%	GPT-2
science	1110	258	+4.3%	372	+3.0%	-44.2%	GPT-2
pt-news	795	282	+2.8%	156	+5.1%	+44.7%	Ours
fwe-train	3,209,616	1,149,595	+2.8%	701,754	+4.6%	+39.0%	Ours
fwe-val	2,661,219	955,342	+2.8%	571,106	+4.7%	+40.2%	Ours

vs. GPT-4 (vocab 100 277)

		GPT-4		Ours			
Text Type	Bytes	Tokens	Ratio	Tokens	Ratio	Rel. Diff %	Winner
news	1813	387	+4.7%	592	+3.1%	-53.0%	GPT-4
korean	877	364	+2.4%	844	+1.0%	-131.9%	GPT-4
code	1259	309	+4.1%	672	+1.9%	-117.5%	GPT-4
math	1830	831	+2.2%	1107	+1.6%	-33.2%	GPT-4
science	1110	249	+4.5%	372	+3.0%	-49.4%	GPT-4
pt-news	795	215	+3.7%	156	+5.1%	+27.4%	Ours
fwe-train	3,209,616	883,058	+3.6%	701,754	+4.6%	+20.5%	Ours
fwe-val	2,661,219	729,099	+3.6%	571,106	+4.7%	+21.7%	Ours